

Chapter 5 Solving Maxwell's Equations

Maxwell's equations form a set of partial differential equations that govern all electromagnetic phenomena in the classical regime. One of the most remarkable achievements of these equations is their prediction that light, a long-standing mystery in scientific history, is an electromagnetic wave. This discovery marks one of the greatest triumphs in science.

However, solving Maxwell's equations in their full generality is a complex task. In certain static cases, where the electric field $\vec{E}$ and the magnetic field $\vec{B}$ are decoupled, solutions are relatively straightforward. However, when the charge density ρ and current density $\vec{J}$ are time-varying, $\vec{E}$ and $\vec{B}$ become intricately coupled, making the analysis significantly more complex. In scenarios where the fields change slowly over time, quasi-static approximations can be employed to simplify the problem. These models neglect the time-dependent effects of electromagnetic wave propagation and directly apply electrostatic and magnetostatic approximations. Such approximations are valuable for addressing a wide range of practical problems.

Before delving into the detailed solutions of Maxwell's equations, particularly with respect to wave phenomena, it is essential to first explore the conditions under which quasi-static approximations can be appropriately applied to simplify electromagnetic models.

1. Conditions for Quasi-Static Analysis

Consider a parallel plate capacitor being charged by a time-varying voltage. Our approach is to employ harmonic analysis, decomposing the time-dependence into harmonic functions, to determine the frequencies at which the system can be approximated using quasi-static models. The following figure illustrates the setup, showing the dimensions of the capacitor and the positive directions of the electric and magnetic fields.

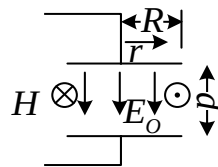


Figure. A plate capacitor being charged by a time-varying voltage.

We assume that the primary electric field is evenly distributed between the metal plates. The harmonic temporal variance of the electric field induces a harmonically-varying magnetic field,

which in turn may generate a secondary electric field. We denote the primary electric field using the complex number $\tilde{E}_0$ which oscillates at an angular frequency ω ($\tilde{E}_0 \propto e^{j\omega t}$), with the uniform distribution of the primary field making $\tilde{E}_0$ independent of spatial coordinates.

Performing a line integral along a circular Amperian loop P in a plane parallel to the metal plates, with a radius r centered around the axis of the capacitor, yields:

$$\oint_P \tilde{H}_{\text{ind}} \cdot d\vec{\ell} = \varepsilon_0 \int_S \partial_t \tilde{E}_0 \cdot d\vec{a} \Rightarrow 2\pi r \tilde{H}_{\phi, \text{ind}}(r) = j\omega \varepsilon_0 \tilde{E}_0 \pi r^2,$$

and we readily obtain:

$$\tilde{H}_{\phi, \text{ind}}(r) = \frac{j\omega \varepsilon_0 r}{2} \tilde{E}_0.$$

The induced magnetic field is in the $\hat{\phi}$ direction. This induced Magnetic field also oscillates at a frequency ω , and it further induces an electric field due to Faraday's law of induction. To evaluate the induced (secondary) electric field, we perform a line integral along the loop (dashed line) shown in the figure below.

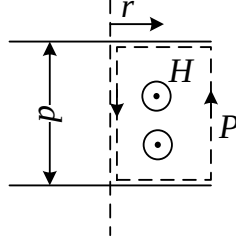


Figure. Integral loop (dashed line) used for calculating the secondary electric field.

Applying Faraday's law in integral form:

$$\oint_P \tilde{E}_0 \cdot d\vec{\ell} = - \int_S \mu_0 \partial_t \tilde{H}_{\text{ind}} \cdot d\vec{a},$$

Because the induced electric field has no radial components due to symmetry constraints, only the axial component (along z) of the electric field at $r = 0$ and $r = R$ contributes to the integral:

$$d \cdot (\tilde{E}(0) - \tilde{E}(R)) = -j\omega \mu_0 \int_0^R \left(\frac{j\omega \varepsilon_0 r}{2} \right) \tilde{E}_0 dr \cdot d,$$

which leads to:

$$\tilde{E}(R) = \tilde{E}(0) - \frac{\varepsilon_0 \mu_0 \omega^2 R^2}{4} \tilde{E}_0.$$

Note that the induced secondary field should be proportional to $\tilde{E}_0$, consequently, $\tilde{E}(0) = \tilde{E}_0$,

and we obtain:

$$\tilde{E}(R) = \tilde{E}_0 - \frac{\varepsilon_0 \mu_0 \omega^2 R^2}{4} \tilde{E}_0.$$

Soon, we'll learn that $\epsilon_0\mu_0 = \frac{1}{c^2}$, $k = \frac{\omega}{c} = \frac{2\pi}{\lambda}$, where c is the speed of light, k is the wavenumber and λ is the wavelength. Using these parameters, the above equation can be written as (recall that R characterizes the size of the capacitor):

$$\begin{aligned}\tilde{E}(R) &= \tilde{E}_0 \left(1 - \frac{1}{4} \left(\frac{\omega}{c} \right)^2 R^2 \right) \\ &= \tilde{E}_0 \left(1 - \frac{1}{4} k^2 R^2 \right) \\ &= \tilde{E}_0 \left(1 - \left(\frac{\pi R}{\lambda} \right)^2 \right).\end{aligned}$$

From the above equation, we observe that as long as $\pi R \ll \lambda$, $\tilde{E}(r) \approx \tilde{E}_0$ everywhere in the capacitor. In conclusion, as long as the dimension of the device is much less than the wavelength, the secondary fields can be neglected. Under such conditions, we can use electrostatics and magnetostatics to solve such quasi-static problems.

Example 1: A power line has a length of $L = 200 \text{ km}$. The frequency of the electromagnetic fields is 60 Hz. Accordingly, we can compute the wavelength of the corresponding electromagnetic wave:

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8 \text{ m/s}}{60/\text{s}} = 5000 \text{ km}.$$

Since $L \ll \lambda$, we can apply quasi-static models to analyze the power line.

Example 2: A MOSFET transistor has a width $w = 100 \text{ nm}$ and it operates at 3 GHz. We can compute the corresponding wavelength as:

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8 \text{ m/s}}{3 \times 10^9/\text{s}} = 0.1 \text{ m}.$$

Since $w \ll \lambda$, the quasi-static condition is met.

Alternatively, if the quasi-static condition is not satisfied, we must rigorously solve Maxwell's equations when dealing with dynamic fields.

2. General solution to Maxwell's equations

1) Expressing the fields using potentials

Maxwell's equations are listed below:

$$\nabla \cdot \vec{E} = \rho / \epsilon_0$$

$$\nabla \times \vec{E} = -\partial_t \vec{B}$$

$$\nabla \cdot \vec{B} = 0$$

$$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \partial_t \vec{E}$$

Note that $\vec{E}$ and $\vec{B}$ are both functions of space and time, denoted as $\vec{E}(\vec{r}, t)$, $\vec{B}(\vec{r}, t)$. They are supposed to be generated by dynamic distributions of $\vec{J}(\vec{r}, t)$ and $\rho(\vec{r}, t)$. In the previous Chapter, we have learnt that the fields can be expressed by the scalar and vector potentials:

$$\vec{E} = -\nabla\phi - \partial_t \vec{A},$$

$$\vec{B} = \nabla \times \vec{A}.$$

Note that the above expressions automatically fulfill:

$$\nabla \cdot \vec{B} = 0,$$

$$\nabla \times \vec{E} = -\partial_t \vec{B}.$$

Now we insert them into Gauss's Law and Ampere's Law to get two new equations:

Inserting $\vec{E} = -\nabla\phi - \partial_t \vec{A}$ into $\nabla \cdot \vec{E} = \rho/\epsilon_0$, we get:

$$\nabla^2 \phi + \partial_t \nabla \cdot \vec{A} = -\rho/\epsilon_0, \quad (1)$$

Inserting the potential expressions into Ampere's Law leads to:

$$\nabla \times \vec{B} = \nabla \times (\nabla \times \vec{A}) = \mu_0 \vec{J} + \mu_0 \epsilon_0 \partial_t (-\nabla\phi - \partial_t \vec{A}),$$

$$\nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A} = \mu_0 \vec{J} - \mu_0 \epsilon_0 \nabla(\partial_t \phi) - \mu_0 \epsilon_0 \partial_{tt} \vec{A},$$

and finally,

$$(\nabla^2 \vec{A} - \mu_0 \epsilon_0 \partial_{tt} \vec{A}) - \nabla(\nabla \cdot \vec{A} + \mu_0 \epsilon_0 \partial_t \phi) = -\mu_0 \vec{J}. \quad (2)$$

Note that equations (1) and (2) include all information about the electromagnetic fields. Moreover, they somehow reduce the complexity of the problem—the number of unknowns is reduced from 6 to 4.

2) Gauge transformations

Suppose potentials ϕ_1 and $\vec{A}_1$ satisfy $\vec{E} = -\nabla\phi_1 - \partial_t \vec{A}_1$, $\vec{B} = \nabla \times \vec{A}_1$. Given an arbitrary scalar function, $\lambda(x, y, z, t)$, and performing the following transformation:

$$\vec{A}_2 = \vec{A}_1 + \nabla\lambda,$$

while at the same time,

$$\phi_2 = \phi_1 - \partial_t \lambda.$$

We can easily prove that $\vec{E}$ and $\vec{B}$ are unchanged under the above transformation:

$$\vec{B} = \nabla \times \vec{A}_2 = \nabla \times (\vec{A}_1 + \nabla \lambda) = \nabla \times \vec{A}_1,$$

$$\vec{E} = -\nabla \phi_2 - \partial_t \vec{A}_2 = -\nabla \phi_1 + \partial_t \nabla \lambda - \partial_t \vec{A}_1 - \partial_t \nabla \lambda = -\nabla \phi_1 - \partial_t \vec{A}_1.$$

It is shown that the choice of ϕ and $\vec{A}$ has certain ambiguity. The transformation from $\{\phi_1, \vec{A}_1\}$ to $\{\phi_2, \vec{A}_2\}$ is called gauge transformation.

Note:

Now it's time for us to pause for a while to check the structure of Maxwell's equations. In Maxwell's equations, there are six unknowns, namely, $E_x, E_y, E_z, B_x, B_y, B_z$, but there are eight equations: two from the divergence equations:

$$\partial_x E_x + \partial_y E_y + \partial_z E_z = \rho / \epsilon_0;$$

$$\partial_x B_x + \partial_y B_y + \partial_z B_z = 0,$$

and six due to the curl equations:

$$\begin{cases} \partial_y E_z - \partial_z E_y = -\partial_t B_x \\ \partial_z E_x - \partial_x E_z = -\partial_t B_y; \\ \partial_x E_y - \partial_y E_x = -\partial_t B_z \end{cases}$$

$$\begin{cases} \partial_y B_z - \partial_z B_y = \mu_0 J_x + \mu_0 \epsilon_0 \partial_t E_x \\ \partial_z B_x - \partial_x B_z = \mu_0 J_y + \mu_0 \epsilon_0 \partial_t E_y. \\ \partial_x B_y - \partial_y B_x = \mu_0 J_z + \mu_0 \epsilon_0 \partial_t E_z \end{cases}$$

It seems that the number of equations (eight) is greater than that of the unknown variables (six). To investigate further, we take the divergence of the curl equations,

$$\nabla \cdot (\nabla \times \vec{E}) \equiv 0 = -\partial_t \nabla \cdot \vec{B}.$$

The above equation dictates that $\nabla \cdot \vec{B} = f(x, y, z)$ is independent of time. The divergence equation $\nabla \cdot \vec{B} = 0$ set the initial condition that $f(x, y, z) = 0$.

Similarly,

$$\nabla \cdot (\nabla \times \vec{B}) = \mu_0 \nabla \cdot \vec{J} + \epsilon_0 \mu_0 \partial_t (\nabla \cdot \vec{E}) = 0.$$

Substituting the continuity equation $\nabla \cdot \vec{J} + \partial_t \rho = 0$,

$$\partial_t (\nabla \cdot \vec{E}) = \frac{1}{\epsilon_0} \partial_t \rho.$$

Integrating both sides over time, we obtain

$$\nabla \cdot \vec{E} = \frac{1}{\epsilon_0} \rho + g(x, y, z),$$

and Gauss's Law sets the initial condition $g(x, y, z) = 0$.

In conclusion, the divergence equations serve as initial conditions. They fix the irrotational parts of E and B , while the curl equations determine the solenoidal parts and time evolution.

3) Lorenz gauge

Let's rewrite the two equations for potentials:

$$\nabla^2 \phi + \partial_t (\nabla \cdot \vec{A}) = -\rho / \epsilon_0 \quad (1)$$

$$(\nabla^2 \vec{A} - \mu_0 \epsilon_0 \partial_{tt} \vec{A}) - \nabla (\nabla \cdot \vec{A} + \mu_0 \epsilon_0 \partial_t \phi) = -\mu_0 \vec{J} \quad (2)$$

We know there is freedom in choosing $\vec{A}$ and ϕ due to gauge invariance. In magnetostatics, we often set

$$\nabla \cdot \vec{A} = 0.$$

This choice simplifies equation (1), but it complicates equation (2). Alternatively, we can simplify equation (2) by setting

$$\boxed{\nabla \cdot \vec{A} + \mu_0 \epsilon_0 \partial_t \phi = 0}.$$

Under this condition, equation (2) reduces significantly to:

$$\boxed{\nabla^2 \vec{A} - \mu_0 \epsilon_0 \partial_{tt} \vec{A} = -\mu_0 \vec{J}}.$$

However, we must first verify that setting $\nabla \cdot \vec{A} + \mu_0 \epsilon_0 \partial_t \phi = 0$ is a valid choice.

Proof:

Suppose initially $\nabla \cdot \vec{A} + \mu_0 \epsilon_0 \partial_t \phi \neq 0$, so we assume $\nabla \cdot \vec{A} + \mu_0 \epsilon_0 \partial_t \phi = \xi(\vec{r}, t)$. Let's perform a gauge transformation:

$$\begin{aligned} \vec{A} + \nabla \lambda &\rightarrow \vec{A}' \\ \phi - \partial_t \lambda &\rightarrow \phi' \end{aligned}$$

Assume that after the transformation, the new potentials $\vec{A}'$ and ϕ' satisfy

$$\nabla \cdot \vec{A}' + \mu_0 \epsilon_0 \partial_t \phi' = 0,$$

which leads to

$$\nabla \cdot \vec{A} + \nabla^2 \lambda + \mu_0 \epsilon_0 \partial_t \phi - \mu_0 \epsilon_0 \partial_{tt} \lambda = 0.$$

Substituting $\nabla \cdot \vec{A} + \mu_0 \epsilon_0 \partial_t \phi = \xi(\vec{r}, t)$ into the above equation, we obtain

$$\nabla^2 \lambda - \mu_0 \epsilon_0 \partial_{tt} \lambda = -\xi(\vec{r}, t).$$

This is a standard inhomogeneous wave equation, and a solution is guaranteed to exist. Thus, we can always perform a gauge transformation to ensure that $\nabla \cdot \vec{A} + \mu_0 \epsilon_0 \partial_t \phi = 0$.

Substituting $\nabla \cdot \vec{A} + \mu_0 \epsilon_0 \partial_t \phi = 0$ into (1), we obtain

$$\boxed{\nabla^2 \phi - \mu_0 \epsilon_0 \partial_{tt} \phi = -\frac{\rho}{\epsilon_0}}.$$

Now the equations for the scalar potential ϕ and the vector potential $\vec{A}$ are both inhomogeneous wave equations. This particular gauge, in which $\nabla \cdot \vec{A} + \mu_0 \epsilon_0 \partial_t \phi = 0$ (as opposed to Coulomb's

gauge, in which $\nabla \cdot \vec{A} = 0$) is called Lorenz gauge.

Note:

From the above two equations, ρ and $\vec{J}$ seems to be independent, i.e.:

$$\nabla^2 \phi - \mu_0 \epsilon_0 \partial_{tt} \phi = -\frac{\rho}{\epsilon_0} \quad (3)$$

$$\nabla^2 \vec{A} - \mu_0 \epsilon_0 \partial_{tt} \vec{A} = -\mu_0 \vec{J} \quad (4)$$

However, according to the continuity equation, ρ and $\vec{J}$ are related through $\nabla \cdot \vec{J} = -\partial_t \rho$.

Now let's explain this discrepancy.

Applying divergence to equation (4), we obtain:

$$\nabla \cdot \nabla^2 \vec{A} - \mu_0 \epsilon_0 \nabla \cdot (\partial_{tt} \vec{A}) = -\mu_0 \nabla \cdot \vec{J}.$$

Since ∇^2 is a scalar operator, we can change the order of $\nabla \cdot$ and ∇^2 to get:

$$\nabla^2 (\nabla \cdot \vec{A}) - \mu_0 \epsilon_0 \partial_{tt} (\nabla \cdot \vec{A}) = -\mu_0 \nabla \cdot \vec{J}.$$

We can change the order because t and $\vec{r}$ are different independent variables.

Applying time-derivative to equation (3),

$$\nabla^2 (\partial_t \phi) - \mu_0 \epsilon_0 \partial_{tt} \phi = -\frac{1}{\epsilon_0} \frac{\partial}{\partial t} \rho.$$

From the above two equations, it is straightforward to obtain:

$$\nabla^2 \left(\frac{1}{\mu_0} (\nabla \cdot \vec{A} + \mu_0 \epsilon_0 \partial_t \phi) \right) - \epsilon_0 (\partial_{tt} (\nabla \cdot \vec{A} + \mu_0 \epsilon_0 \partial_t \phi)) = -(\nabla \cdot \vec{J} + \partial_t \rho).$$

The condition for Lorenz transformation $\nabla \cdot \vec{A} + \mu_0 \epsilon_0 \partial_t \phi = 0$ guarantees that

$$\nabla \cdot \vec{J} + \partial_t \rho = 0.$$

4) EM waves

Equations (3) and (4) are both inhomogeneous wave equations. We define the d'Alembert operator (There is some variation in the symbolic notation for this operator. While it is often denoted simply by the box symbol " $\square$ ", here we follow Griffith's convention, which uses a superscripted box " $\square^2$ "):

$$\square^2 \equiv \nabla^2 - \mu_0 \epsilon_0 \partial_{tt},$$

to simplify the equations as:

$$\square^2 \phi = -\frac{1}{\epsilon_0} \rho,$$

$$\square^2 \vec{A} = \mu_0 \vec{J}.$$

In source-free regions where ρ and $\vec{J}$ are both zero, the potentials satisfy the homogeneous wave

equations:

$$\square^2 \phi = 0,$$

$$\square^2 \vec{A} = 0.$$

Later, we will see that the solutions to these equations take the form of waves. Since the electric and magnetic fields can be expressed in terms of the potentials, it follows that the fields themselves must also be wave-like. We will explicitly demonstrate this below.

We apply the curl operator to equation (4) and get:

$$\nabla^2(\nabla \times \vec{A}) - \mu_0 \epsilon_0 \partial_{tt}(\nabla \times \vec{A}) = -\mu_0 \nabla \times \vec{J}.$$

Recognizing that $\nabla \times \vec{A} = \vec{B}$, this becomes the wave equation for the magnetic field:

$$\boxed{\square^2 \vec{B} = -\mu_0 \nabla \times \vec{J}}.$$

For the electric field, we start with $\vec{E} = -\nabla\phi - \partial_t \vec{A}$ and compute its wave equation:

$$\square^2 \vec{E} = \nabla^2(-\nabla\phi - \partial_t \vec{A}) - \mu_0 \epsilon_0 \partial_{tt}(-\nabla\phi - \partial_t \vec{A}).$$

Expanding and substituting the Lorenz gauge conditions $\nabla^2\phi = \mu_0 \epsilon_0 \partial_{tt}\phi - \rho/\epsilon_0$ and $\nabla^2 \vec{A} = \mu_0 \epsilon_0 \partial_{tt} \vec{A} - \mu_0 \vec{J}$,

$$\square^2 \vec{E} = -\nabla \underbrace{(\nabla^2 \phi)}_{\mu_0 \epsilon_0 \partial_{tt} \phi - \rho/\epsilon_0} - \partial_t \underbrace{(\nabla^2 \vec{A})}_{\mu_0 \epsilon_0 \partial_{tt} \vec{A} - \mu_0 \vec{J}} + \mu_0 \epsilon_0 \partial_{tt}(\nabla\phi) + \mu_0 \epsilon_0 \partial_{ttt} \vec{A}.$$

After cancellations, this simplifies to the wave equation for the electric field:

$$\boxed{\square^2 \vec{E} = \nabla \left(\frac{\rho}{\epsilon_0} \right) + \mu_0 \frac{\partial}{\partial t} \vec{J}}$$

In fact, the wave equations for $\vec{E}$ and $\vec{B}$ follow directly from Maxwell's equations. For instance, taking the curl of the Maxwell-Faraday equation and then incorporating Ampère's Law (with Maxwell's correction) and Gauss's Law yields the wave equation above.

In free space where ρ and $\vec{J}$ are both zero, $\vec{E}$ and $\vec{B}$ satisfy the homogeneous wave equations:

$$\square^2 \vec{E} = 0,$$

$$\square^2 \vec{B} = 0.$$

In the following sections, we will solve these equations step by step. First, we examine the homogeneous wave equation, whose general solution describes wave propagation in source-free regions. Finally, we address the radiation problem by solving the inhomogeneous wave equation, where sources are present. Before proceeding with the solutions, we first outline the general

structure of wave equation solutions, using the scalar potential ϕ and vector potential $\vec{A}$ as illustrative example. For any inhomogeneous differential equation, the complete solution consists of two distinct components:

Part 1: Particular solution: A solution that satisfies the inhomogeneous equation with the source terms. Let's call these ϕ_p and $\vec{A}_p$.

Part 2: Homogeneous solution: A solution to the corresponding homogeneous equation (where the right-hand side is zero). Let's call these ϕ_h and $\vec{A}_h$.

So, mathematically, the full solution is:

$$\phi = \phi_p + \phi_h, \quad \vec{A} = \vec{A}_p + \vec{A}_h.$$

For the radiation problem to be addressed in the final section of this chapter, the solutions take the form of retarded potentials, which represent particular solutions exclusively—no homogeneous component contributes in this case. This exclusive appearance of the particular solution stems from boundary condition considerations, as will be examined in detail in the corresponding discussion.

5) Homogeneous Solutions

Denote $\mu_0 \epsilon_0 = \frac{1}{c^2}$ where c is the wave speed (as derived later). The homogeneous solutions satisfy the wave equation:

$$\square^2 f = 0,$$

where $\square^2 \equiv \nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}$. For a vector field such as $\vec{A}$, each of its three components satisfies the scalar wave equation above. We begin by analyzing the simplest case - the one-dimensional wave equation - which serves as a particularly illustrative example.

① One-dimensional wave equations:

In 1D, the homogeneous wave equation becomes:

$$\partial_{zz} f = \frac{1}{c^2} \partial_{tt} f.$$

Below, we will prove that the solution to the above equation is in the form of:

$$f(z, t) = g(\underbrace{z - ct}_u),$$

or:

$$f(z, t) = h(z + ct).$$

Proof: we will prove that $g(z - ct)$ is a solution. The case for $h(z + ct)$ follows analogously.

$$\begin{aligned}
\partial_{zz}f &= \partial_z(\partial_z f) = \partial_z \left(\frac{dg}{du} \cdot \frac{du}{dz} \right) = \partial_z \left(\frac{dg}{du} \right) \\
&= \left(\frac{d^2g}{du^2} \right) \frac{du}{dz} = \frac{d^2g}{du^2}. \\
\partial_{tt}f &= \partial_t(\partial_t f) = \partial_t \left(\frac{dg}{du} \cdot \frac{du}{dt} \right) = \partial_t \left(\frac{dg}{du} \cdot (-c) \right) \\
&= \left(\frac{d^2g}{du^2} \right) (-c) \cdot \frac{du}{dt} = c^2 \frac{d^2g}{du^2} = c^2 \partial_{zz}f.
\end{aligned}$$

From the above equations, we conclude that:

$$\partial_{zz}f = \frac{1}{c^2} \partial_{tt}f.$$

For example, if g is a Gaussian function, $f(z, t) = e^{-(z-ct)^2}$ retains its Gaussian shape and propagates in the $+z$ direction, as illustrated in the following figure.

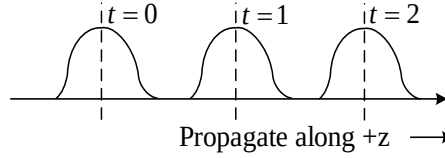


Figure. A wave in the form of $e^{-(z-ct)^2}$ propagates along $+z$.

Similarly, the wave $f(z, t) = e^{-(z+ct)^2}$ has the same shape but propagates along $-z$.

A commonly-used form of g is $g(u) = A_0 \cos(ku + \phi_0)$, where A_0 , k and ϕ_0 are constants.

Inserting $u = z - ct$, we get:

$$\begin{aligned}
f(z, t) &= A_0 \cos(k(z - ct) + \phi_0) \\
&= A_0 \cos(\underbrace{kz - kct + \phi_0}_{\text{total phase}})
\end{aligned}$$

All terms in " " have a unit of [rad]. So k has a unit of $\frac{[rad]}{[m]}$. As a scalar quantity, it is termed the wavenumber. When the total phase increases by 2π , the cosine pattern will repeat itself. By setting $kz_0 = 2\pi$, we get the periodicity of the spatial pattern: $z_0 = \frac{2\pi}{k}$, while in the time domain, $kct = 2\pi$ yields a period of $T = \frac{2\pi}{kc}$. Here, we denote $kc = \omega$ as the angular frequency.

It is critical to emphasize that the solution described above, while theoretically elegant, is practically unrealistic. This dichotomy arises because the cosine function inherently extends infinitely in both space and time—a mathematical idealization that cannot physically manifest. Nevertheless, its utility lies in its role as a foundational basis for Fourier analysis, wherein nearly any function can be synthesized through superposition of such harmonic components.

Furthermore, operations involving cosine functions can be significantly simplified by employing complex exponential notation. For any linear operator $\hat{L}$ acting on a function $e^{j\psi}$ (ψ is a function that $\hat{L}$ acts on, for example, $\psi = kz - kct + \phi_0$, while $\hat{L} = \frac{\partial}{\partial z}$), the following identity holds:

$$\text{Re}\{\hat{L}e^{j\psi}\} = \hat{L}\text{Re}\{e^{j\psi}\} = \hat{L}\cos\psi.$$

This equivalence arises directly from Euler's formula $e^{j\psi} = \cos\psi + j\sin\psi$ and the linearity of the operator. Crucially, working with the complex exponential $e^{j\psi}$ proves far more tractable than manipulating trigonometric functions directly. In electromagnetic wave analysis, this approach is particularly valuable - calculations are typically performed using the complex exponential form, with physical quantities subsequently obtained by taking the real part ($\text{Re}\{\}$) of the final result. Now we extend our discussion to three-dimensions, which is more relevant to realistic EM waves.

② Three-dimensional wave equations:

In three dimensions, we generalize the harmonic solution from the 1D case to construct the wavefunction:

$$f(\vec{r}, t) = A_0 e^{j(-\omega t + \vec{k} \cdot \vec{r} + \phi_0)},$$

where $\vec{k}$ is the wave vector whose magnitude $k = \frac{\omega}{c} = \frac{2\pi}{\lambda}$ represents the wavenumber, and whose direction specifies the wave's propagation orientation in space.

It is important to note that in any plane perpendicular to the propagation direction (defined by the unit vector $\hat{k}$), $\vec{k} \cdot \vec{r} = kr\cos\theta$ remains constant. Consequently, the phase (at a fixed time) is uniform across such planes - as illustrated below. Since these surfaces of constant phase are planar, this harmonic solution is referred to as a plane wave.

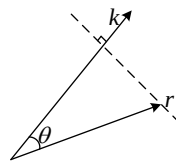


Figure. Across the plane perpendicular to $\hat{k}$, denoted by the dashed line, phase of the wave is constant.

Plane waves are sometimes denoted as

$$f(\vec{r}, t) = \tilde{A}_0 e^{j(-\omega t + \vec{k} \cdot \vec{r})} = \tilde{A}_0 e^{j\vec{k} \cdot \vec{r}} e^{-j\omega t} = R(\vec{r})T(t),$$

In the above notation, the initial phase ϕ_0 is subsumed within the complex amplitude $\tilde{A}_0 = A_0 e^{j\phi_0}$, resulting in a product decomposition where the spatial and temporal dependencies are completely separated. Accordingly, the homogeneous wave equation can be modified:

$$\left(\nabla^2 - \frac{1}{c^2} \partial_{tt} \right) R(\vec{r}) T(t) = 0,$$

$$\Rightarrow (\nabla^2 R(\vec{r})) T(t) = \frac{1}{c^2} R(\vec{r}) \partial_{tt} T(t).$$

This reminds us of the treatment in separation of variables:

$$\frac{\nabla^2 R(\vec{r})}{R(\vec{r})} = \frac{1}{c^2} \frac{\partial_{tt} T(t)}{T(t)}.$$

Despite the fact that we arrive at the above equation starting from plane waves, it should be noted that for any function $u(\vec{r})$, as long as it can be expressed as $u(\vec{r}) = R(\vec{r}) T(t)$, separation of variable can be applied. For example, for a single-frequency wave (having an angular frequency ω),

$$f(\vec{r}, t) = \tilde{A}_0(\vec{r}) e^{-j\omega t},$$

For the temporal part,

$$\frac{1}{c^2} \frac{\partial_{tt} T(t)}{T(t)} = -k^2,$$

where k is the wavenumber, and the spatial part must obey

$$\frac{\nabla^2 R(\vec{r})}{R(\vec{r})} = -k^2.$$

The above equation translates into

$$\boxed{\nabla^2 R(\vec{r}) + k^2 R(\vec{r}) = 0},$$

which is termed the Helmholtz equation. In the next chapter, we will examine wave phenomena - including reflection, transmission, waveguiding, and diffraction - within the framework of monochromatic fields. Throughout these analyses, we will frequently work with solutions to the Helmholtz equation to characterize these single-frequency wave behaviors.

It is important to note that the three-dimensional wave equation admits diverse classes of solutions, depending on the coordinate system employed. In spherical coordinates, for instance, the simplest solution—exhibiting no angular dependence (i.e., independent of θ and ϕ)—takes the form of a spherical wave. This solution can be expressed as:

$$f(|\vec{r}|, t) = \frac{C_1}{|\vec{r}|} g(|\vec{r}| - ct) + \frac{C_2}{|\vec{r}|} g(|\vec{r}| + ct).$$

The first term on the right-hand side corresponds to a diverging spherical wave, while the second term describes a converging spherical wave.